

On the QUD sensitivity of a third reading

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Introduction Recent implementations of substitution theories of transparent readings in attitude reports (Percus, 2020; Benbaji-Elhadad, 2023; Mayr and Schmitt, 2024) argue that the availability of a transparent reading is influenced by the Question Under Discussion (QUD). Although claims of QUD-sensitivity are often supported with experimental evidence (e.g. Tonhauser et al. 2018; Champollion et al. 2019), existing experimental work on context-sensitivity of transparent readings (Zhang and Davidson, 2021, 2024) does not directly manipulate QUDs, a gap that this study aims to fill. In particular, our study investigates availability of the transparent reading of the descriptive content of an indefinite DP (i.e., the ‘third’ readings; Fodor 1970) under two different QUDs about the attitude holder’s doxastic state, and the results provide evidence for the introspective judgments provided by Mayr and Schmitt (2024), which pose challenges for some implementations of the QUD-sensitive replacement approach (Percus, 2020).

Background (1) is true in the context provided, even though Olivia does not know the women in question to be fitness coaches. The true reading of (1) is a ‘third reading’, where *fitness coaches* is interpreted transparently while the existential quantifier of the indefinite scopes under *think*.

- (1) Context: Looking at a group of women, Olivia thinks her brother is seeing one of them, although she has no idea which one. Unbeknownst to her, the women are fitness coaches. Olivia thinks that her brother is seeing a *fitness coach*.

Under one approach to transparent readings, every descriptive content in the embedded clause is evaluated at the attitude holder’s doxastic alternatives, but the *res* (Schwager, 2009; Sudo, 2014; Mayr and Schmitt, 2024) or the entire embedded proposition (Percus, 2020; Benbaji-Elhadad, 2023) is replaced by something that the attitude holder in fact has a *de dicto* belief about. A subset of such replacement-based theories refer to the QUD in their definedness conditions (Percus, 2020; Benbaji-Elhadad, 2023; Mayr and Schmitt, 2024). Percus (2020) postulates a covert operator R (2) that replaces the embedded proposition with a alternative proposition (p_8). Crucially, such replacement is constrained in that the prejacent proposition is required to address the QUD (i.e., is at-issue in c), and to be contextually entailed by the replacing proposition.

- (2) $\llbracket \text{Olivia thinks that R } p_8 \text{ her brother is seeing a fitness coach} \rrbracket^{g,w,c} = \forall w' \in \text{DOX}_{O,w}: p(w')$
where $p = \lambda w. \text{ Olivia's brother is seeing one of the women she saw in } w$, $q = \lambda w. \text{ Olivia's brother is seeing a fitness coach in } w$, defined only if q is at issue in c & $\forall w' \in c: p(w') \rightarrow q(w')$

In contrast, for Mayr and Schmitt, what must address the QUD is not the embedded proposition but the entire attitude report. (1) is defined iff the proposition that Olivia thinks her brother is seeing one of the women she saw (read *de dicto*) entails the same answer to the QUD as the proposition that Olivia thinks her brother is seeing a fitness coach (read *de dicto*). What motivates Mayr and Schmitt to refer to at-issueness of the entire attitude report is that, among QUDs about the attitude holder’s doxastic state, some make a transparent reading available while others do not. We investigate the predictions of these theories by directly manipulating the QUD.

Experiment Our experiment involved a 2×2 design crossing QUD with transparency of the descriptive content of an indefinite noun phrase. One type of QUD (henceforth ‘existential questions’) asks whether the attitude holder believes that there is an individual with a certain property (regardless of how they are represented in their mind), e.g. in (1), an existential question would be *Does Olivia think that her brother is in love?* The other type (henceforth ‘*wh* questions’) inquires about the attitude holder’s belief about the *res*, e.g. in (1), it would be *What does Olivia think the person her brother is seeing does for work?* Participants see a story followed by target and filler statements, and rate each statement given the story on a five-point Likert scale ranging from ‘Highly Agree’ to ‘Highly Disagree’. The QUD is set up by an embedded question under *wonder* in a story, and repeated as the first sentence of a target item.

(3) and (4) present stimuli for a transparent reading under an existential QUD and a *wh* QUD, respectively. Note that the attitude report in the target item is only true under the transparent reading of *programmer*. The attitude report remains string identical across the four conditions.

(3) transparent reading $\times$ existential QUD

- a. Story: Mia moved into a new apartment last week. *Her floor is so quiet, so she is wondering if a studio apartment next door is vacant.* One day, Mia sees a group of men in the hallway. The men are programmers, but Mia doesn't know that. Mia thinks that one of the men lives in the apartment next door, although she cannot tell which one. One of them indeed lives in the apartment next door.
- b. Target: *Mia was wondering if the apartment next door is vacant.* After the encounter in the hallway, she thinks that a programmer lives next door.

(4) transparent reading $\times$ *wh* QUD

- a. Story: Mia moved into a new apartment last week. *She knows that a studio apartment next door is occupied, but she knows nothing about the person who lives there. Mia wants to know about her neighbor. She is especially wondering what her next door neighbor does for work.* One day, Mia sees a group of men in the hallway. The men are programmers, but Mia doesn't know that. Mia thinks that one of the men lives in the apartment next door, although she cannot tell which one. One of them indeed lives in the apartment next door.
- b. Target: *Mia was wondering what her next door neighbor does for work.* After the encounter in the hallway, she thinks that a programmer lives next door.

QUD and transparency were tested within-subjects in a Latin-square design, where each participant saw each of the four conditions exactly once, presented in different stories. Participants (30 adult native speakers of English) were recruited through Prolific. After applying exclusion criteria for attention based on catch trials, data from 28 participants were included in analysis.

Predictions and Results Percus (2020) predicts that a transparent reading cannot be derived under either QUD because they make the entire attitude report rather than the embedded proposition at-issue. In contrast, Mayr and Schmitt predict that a transparent reading is available under existential questions but not under *wh* questions; whether Mia identifies a next door neighbor as a programmer or not, she believes that the apartment is not vacant, but whether Mia identifies the neighbor as a programmer does change the answer to what she thinks he does for work.

We fit Bayesian multilevel cumulative ordinal models (rating $\sim$ QUD*transparency + story + (1+ QUD*transparency|subject)), with the existential question and the transparent condition as the reference levels for QUD and transparency, respectively. The main effect of transparency is significant ($\beta=7.56$, HPD=[2.35, 13.78]). There is a marginal effect of the QUD such that *wh* QUDs are associated with lower acceptability ratings for transparent readings ($\beta=-2.52$, CI=[-6.25, -0.17], HPD=[-5.93, 0.01], Fig. 1). Our results thus provide tentative support for Mayr and Schmitt's claim of a QUD effect, although the effect is far from categorical. A closer investigation shows that the facilitation effect of an existential question is more robust for a subset of participants who chose "Highly Agree" for a transparent reading at least once (Fig. 2).

Conclusion We provide what we believe is the first quantitative evidence for QUD-sensitivity of third readings, supporting Mayr and Schmitt's empirical claim that even among QUDs about the attitude holder's doxastic state, there is difference in whether they allow a transparent reading of an attitude report. At the same time, the overall preference for opaque readings reported in previous studies (Zhang and Davidson, 2021, 2024) is replicated, and the effect of the QUD may vary by whether a speaker accesses a transparent reading in the first place. To the extent that transparent readings are available under existential questions, replacement theories that require at-issueness of the embedded proposition (Percus, 2020) seem to under-generate such readings,

in contrast to replacement theories based on evaluation of the entire attitude report.

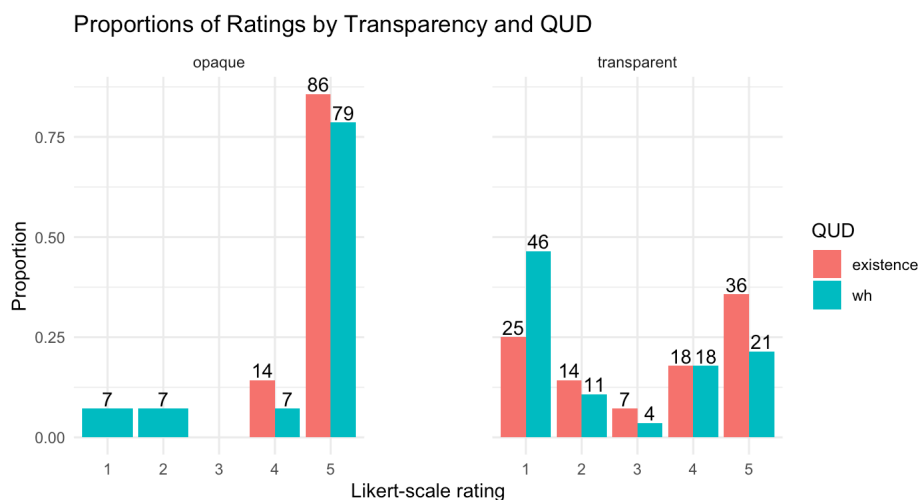


Figure 1: Percentage of Likert-scale responses by QUD in opaque (left) and transparent (right) conditions

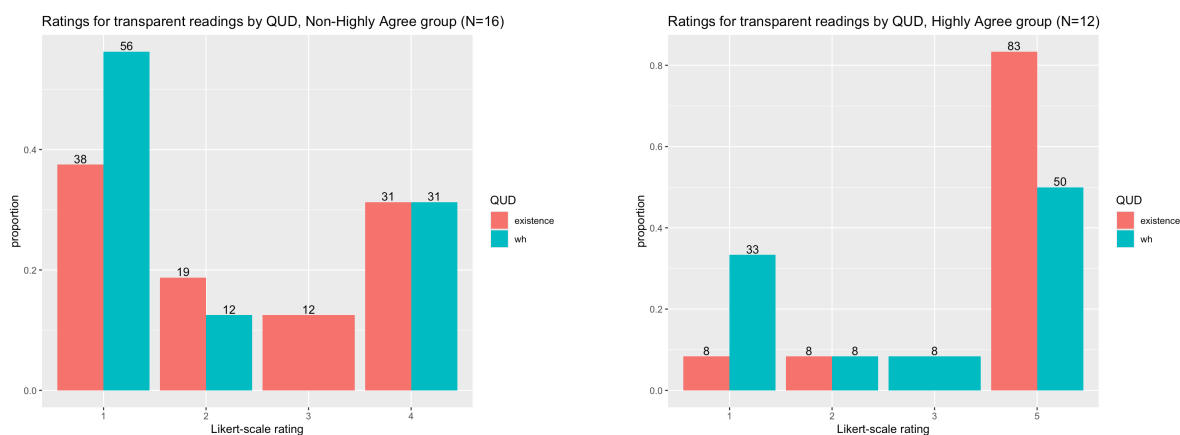


Figure 2: Ratings for transparent readings by participants who never chose “Highly Agree” with a transparent reading (left) and those who did so at least once (right)

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